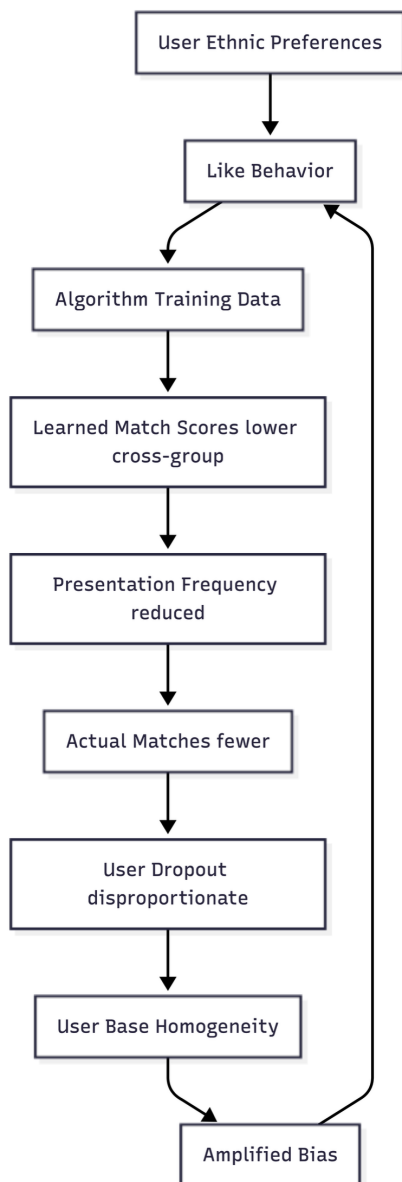


MADS Ethics Portfolio assignment 2 – Breeze dating app

First impressions

The core issue is that the self-learning algorithm amplifies existing social biases in romantic preference. Even if there is a mild preference, this mild preference will be amplified. The model learns that it receives a lower score by recommending people with a different ethnic background. This turns in a self-fulfilling prophecy where in each learning rate less people of different background are presented. In the end this will lead to a complete removal of this category.

The DAG



Missed insights revealed after DAG

I missed the fact that users are not only presented less but since they get fewer matches, they will leave the app. This will make the userbase become even more homogeneous. This will change the training data permanently and makes it much harder to correct the bias later.

Advise for the data scientist

The problem is not always maximizing the number of matches. This example shows that minority groups are leaving the app making it less attractive for all users. Furthermore, this also leads to legal issues. This is a serious matter and should be addressed immediately.

In order to show what happens to the rest of the team I would propose to create a simple model that shows what happens without intervention.

The system needs to be designed so it learns user preference without amplifying them into discrimination. A post-processing re-ranking should be used to ensure fair representation rates on short notice. In order to fix the problem for the long term fairness constraints can be introduced in the cost function of the model.